Aadhithya Bharathi A

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Education

Amrita Vishwa Vidyapeetham, B.Tech in Computer Science — August 2023 – Present
CGPA: 9.0 / 10

IIT Madras, Diploma in Data Science — 2023 – Present

Velammal Bodhi Campus, Higher Secondary — 2021 – 2023

Edison G. Agoram Memorial School, Secondary — 2020 – 2021

Experience

Trusty Bytes | Full Stack Developer Intern — *May 2025 – June 2025*

- Developed the front-end of a demand-supply mobile application using React Native.
- Built responsive and user-friendly interfaces optimized for both Android and iOS.
- Strengthened skills in mobile UI development, component-based architecture, and cross-platform design.

Extra Curriculars

Editor, ASCII Newsletter — January 2024 – Present

Amrita Vishwa Vidhyapeetham

Contributed to content curation and editing of the technical newsletter, improving readability and engagement among student readers.

Ananda Vikatan | Photo Journalist — August 2024 – August 2025

Published photo stories in one of India's most respected Tamil magazines.

Research Presentations

Paper Presenter – Invente'25, SSN College of Engineering, Chennai — *Sep 2025*

Presented research work titled *"A Multimodal AI Framework for Early Diagnosis of Fibromyalgia using WPI and SSI Scales"* at a National Level Tech Symposium jointly organized by Shiv Nadar University & SSN College of Engineering.

Projects

Smart Cafeteria Management System — github.com
(Full Stack, Microservices, ML Forecasting)

- Built a microservices-based platform using **Go (Gin)**, **React.js**, and **Python (Flask)**, deployed with Docker.
- Developed a **Random Forest** model (81.5% accuracy) for demand forecasting using real-time data.
- Implemented queue management, secure authentication (**JWT + 2FA**), and role-based dashboards.
- Added analytics for food waste tracking and booking optimization.

Tools: Go, React.js, Python, PostgreSQL, Docker

FX Decision Recommendation System for Indian Businesses — github.com
(Business Analytics, Time-Series Forecasting, Risk Modeling, Full Stack)

- Built a full-stack FX decision system to analyze currency exposure and generate hedge/spot recommendations.
- Developed forecasting models using **Facebook Prophet** for short-term exchange rate prediction.
- Implemented risk analytics using **volatility, Z-score anomaly detection, and Value-at-Risk (VaR)**.
- Designed a React dashboard for real-time risk visualization and decision support.

Tools: Python, Flask, Pandas, NumPy, Prophet, Statsmodels, React

AI-Driven Multi-Hazard Disaster Monitoring System github.com
(Embedded Systems, IoT, Machine Learning, Full Stack)

- Built an STM32-based system integrating gas, flame, and vibration sensors for real-time hazard detection.
- Implemented communication protocols (ADC, I2C, UART) with Bluetooth alerts using embedded C.
- Developed a **Random Forest** model and deployed inference via **FastAPI**.
- Created a React dashboard for live alerts and hazard visualization.

Tools: STM32, Embedded C, Python, scikit-learn, FastAPI, React

Innovora | Full-Stack Collaboration Platform github.com
(Full Stack, MERN)

- Developed a platform enabling students and mentors to collaborate, share projects, and publish research.

Tools: MongoDB, Express.js, React, Node.js

Tamil Sentence Compiler github.com
(Compiler Design,ANTLR4, Language Processing)

- Built a compiler front-end for Tamil sentences implementing lexical, syntax, and semantic analysis using ANTLR4.
- Developed both a Java-based compiler pipeline and a browser-based JavaScript implementation with real-time validation.

Tools: Java, ANTLR4, JavaScript, HTML, CSS

Cinema Audience Forecasting *Kaggle Competition (IIT Madras Data Science Program)*
(Time-Series Forecasting, Machine Learning)

- Built a forecasting system to predict daily cinema audience using POS and booking data.
- Performed feature engineering and trained regression-based time-series models.

Tools: Python, pandas, NumPy, scikit-learn

Virtual Experience / Certifications

Deloitte Australia Data Analytics Job Simulation | Forage | Mar 2026

- Completed a data analytics simulation focused on business insights and forensic technology use cases.
- Built an interactive Tableau dashboard to visualize downtime by factory and device type.
- Used Excel to classify data and derive actionable business insights.

Skills

Programming Languages: Python, C, Java, JavaScript
Web Technologies: HTML, CSS, ReactJS
Databases: MySQL, MongoDB
Data & Analytics: Data Analysis, Data Modeling, Data Visualization (Tableau), Spreadsheet Skills (Excel), Log Analysis
Software & Systems: Software Development, Computer Networking, Web Security
Tools & Libraries: scikit-learn, pandas, NumPy, matplotlib
Soft Skills: Communication, Teamwork, Problem Solving, Leadership, Emotional Intelligence, Planning